NAKIB AHMED RAJ

M.Sc. / Ph.D. Applicant · Artificial Intelligence & Natural Language Processing

Mail | Github | LinkedIn | Portfolio

Dhaka, Bangladesh

OVERVIEW

Aspiring Ph.D./M.Sc. researcher in Artificial Intelligence and Natural Language Processing, bridging academic research with production ML engineering. Currently a Junior Software Engineer in MLOps and Data Engineering at Poridhi.io, where I build production-grade ML pipelines, observability systems, and agentic LLM tooling. My research background is in multi-task learning for NLP, and I'm increasingly focused on the design and reliability of agentic systems. Seeking to bring this combination of research grounding and applied systems experience into graduate study.

RESEARCH INTERESTS

My primary research interests lie at the intersection of Natural Language Processing (NLP), Trustworthy AI, and Scalable Machine Learning Systems and AI Agents. Specific areas of focus include:

- Multi-task and transfer learning for low-resource NLP tasks, particularly hate speech, sentiment, and aggression detection in multilingual social media contexts
- Robust and interpretable deep learning architectures for text classification, incorporating Transformer-based models (BERT, RoBERTa), convolutional-recurrent hybrids (CNN-BiLSTM), and multi-head self-attention mechanisms
- MLOps and production-grade machine learning systems: pipeline reproducibility, model drift detection, and large-scale data engineering
- Fairness, bias mitigation, and generalization in neural language models
- Agentic AI and LLM-based systems, focusing on tool orchestration, agent state and context management, execution reliability, and developer-centric agentic workflows

EDUCATION

B.Sc. in Mechatronics and Industrial Engineering *2020 – 2025*

Chittagong University of Engineering and Technology (CUET), Chittagong, Bangladesh

- CGPA: 3.27 / 4.00
- Undergraduate Thesis: "Decoding the Spectrum of Online Harassment: A Neural Approach to Sentiment, Hate and Aggression Identification"
- Relevant Coursework: Artificial Intelligence, Digital Signal Processing, Control Systems, Numerical Methods, Engineering Mathematics

Higher Secondary Certificate (HSC) *2017 – 2019*

Mirpur Cantonment Public School and College, Dhaka

- GPA: 5.00 / 5.00 (Science)

Secondary School Certificate (SSC) *2015 – 2017*

Adamjee Cantonment Public School, Dhaka

- GPA: 5.00 / 5.00

RESEARCH EXPERIENCE

Undergraduate Thesis Research *2024 – 2025*

Department of Mechatronics and Industrial Engineering, CUET

Thesis: *"Decoding the Spectrum of Online Harassment: A Neural Approach to Sentiment, Hate and Aggression Identification"*

Overview:

Conducted a systematic investigation into automated online harassment detection using multi-task deep learning. The work addresses three inter-related classification problems like emotion/sentiment analysis, violence/aggression detection, and hate speech classification within a unified neural framework applied to social media text.

Architectures & Methodologies:

- Designed and implemented a hybrid neural encoder combining a from-scratch Transformer, a multi-scale
- CNN, and a stacked BiLSTM with self-attention pooling, fused with hand-crafted linguistic features (length, polarity, word/char counts) for joint emotion, violence, and hate-speech classification
- Benchmarked the proposed model against five baselines, BERT-base, RoBERTa-base, DeBERTa-base, LSTM-only, and CNN-only under an identical multi-task head configuration
- Designed a multi-task learning (MTL) framework sharing lower encoder layers across all three classification tasks, with task-prefix routing 'EMOTION', 'VIOLENCE', 'HATE' and task-specific residual adapter heads, enabling transfer of cross-task linguistic features and reducing task-specific overfitting
- Built a reproducible end-to-end training pipeline using AdamW with cosine warmup, mixed-precision training, gradient checkpointing, multi-GPU via DataParallel, weighted cross-entropy, and task-balanced sampling
- In progress: attention visualisation (attention rollout) to interpret model decision boundaries and audit for demographic bias

Datasets & Scale:

- Curated and pre-processed balanced corpora across three annotation schemes from public benchmarks, harmonised to approximately 150,000 labelled instances (50K per task) using a custom cleaning pipeline (contraction expansion, emoji demojizing, URL/mention/hashtag removal, concatenated-word splitting, NLTK tokenisation, stopword removal, lemmatisation) parallelised with joblib
- Applied manual class balancing via token-swap/token-drop augmentation, plus in-training synonym replacement (via nlpaug) and mixup interpolation between same-task sample pairs to address class imbalance

Evaluation & Results:

- Evaluated on a held-out 10% stratified test split using Accuracy, Macro-F1, Precision, and Recall per task; calibration quantified via ECE, Brier score, and reliability diagrams
- On the preliminary single-seed run (seed 42): Emotion Accuracy 0.957 / Macro-F1 0.957; Violence Accuracy 0.995 / Macro-F1 0.995; Hate speech Accuracy 0.969 / Macro-F1 0.968, with 224.7M parameters, 5.7 GB peak VRAM, and ~22 s inference on the test split (T4 GPUs)
- In progress: stratified k-fold cross-validation (k=5), Matthew's Correlation Coefficient (MCC) reporting, multi-seed statistical significance (5–10 seeds with mean ± std, 95% CI, and p-values vs strongest baseline), and learning-curve analysis across 20/40/60/80/100% training-data fractions
- In progress: robustness analyses including out-of-domain evaluation on an unseen external hate-speech corpus and adversarial text perturbation under 7 attack types (character substitution, misspellings, word deletion, synonym, punctuation, emoji insertion, slang/obfuscation) to demonstrate model generalisation across writing styles and platform dialects

Research Contributions:

- Proposed a unified annotation taxonomy reconciling disparate labelling conventions across existing hate speech corpora
- Investigated intersectional bias in pre-trained language models by probing representations for protected attributes; findings inform fairness-aware fine-tuning procedures
- Manuscript currently in preparation for submission to an NLP/AI venue (venue TBD)

INDUSTRY & PROFESSIONAL EXPERIENCE

Junior Software Engineer – MLOps, Data Engineering, AI Engineering *January 2025 – Present*
Poridhi.io, Dhaka, Bangladesh

- Architected and deployed scalable real-time and batch data processing pipelines using Apache Spark (distributed DataFrame operations over multi-node clusters), Apache Kafka (event streaming at >50k messages/sec throughput), and Apache Flink (stateful stream processing with exactly-once semantics)
- Engineered production-grade ML systems for churn prediction (gradient-boosted and neural classifier ensembles) and fraud detection (anomaly detection via Isolation Forest + LSTM sequence models), covering the full lifecycle from raw data ingestion to containerised model serving
- Implemented comprehensive model observability frameworks integrating Prometheus (metric scraping), Grafana (real-time dashboards), and EvidentlyAI (statistical drift detection on feature and prediction distributions), reducing undetected model degradation incidents
- Provisioned and managed cloud infrastructure on AWS (EC2, S3, RDS, ECR) using Pulumi Infrastructure-as-Code, ensuring reproducible, version-controlled deployments
- Managed end-to-end experiment tracking, model versioning, and staged promotion using MLflow Model Registry, enabling systematic A/B comparison across >30 experimental runs

Software Engineering Intern – Data & ML Infrastructure *October 2024 – December 2024*
Poridhi.io, Dhaka, Bangladesh

- Designed and maintained automated ETL and ML data preparation workflows using Apache Airflow DAGs, reducing manual pipeline intervention time by ~40%
- Containerised ML training and inference services with Docker, establishing reproducible environments across development, staging, and production
- Developed RESTful model inference APIs using FastAPI, implementing asynchronous request handling and input validation schemas
- Operated within Linux-based CI/CD workflows (GitHub Actions), maintaining code quality through pull-request reviews and automated testing

Industrial Attachment – BITAC, Dhaka
Bangladesh Industrial Technical Assistance Centre

- Completed structured training in industrial engineering processes, manufacturing systems, and automation integration
- Gained practical exposure to electro-mechanical systems and CNC-based manufacturing workflows

SELECTED TECHNICAL PROJECTS

Puku-CLI - Agentic Terminal Harness Engine (Active) *2025-2026*

- Designing and testing a CLI-native agentic framework built around hooks, skills, agents, and sub-agents, with a PUKU.md configuration spec (analogous to CLAUDE.md) governing agent behaviour, MCP integrations, and slash command dispatch
- Work explores harness architecture for composable, inspectable agentic pipelines, focusing on how structured context, tool routing, and inter-agent communication can be made reliable and debuggable at the terminal level

Multi-Task NLP Pipeline for Online Harassment Detection *2024 – 2025*

- End-to-end Python pipeline using HuggingFace Transformers, PyTorch, and Scikit-Learn for data preprocessing, model training, hyperparameter optimisation (Optuna), and evaluation reporting
- Trained on ~80,000 labelled social media instances; achieved macro-F1 of ~0.87 (hate) and ~0.83 (aggression) with fine-tuned BERT-base-uncased
- Experiment tracking via MLflow; results reproducible via Docker containerisation

Production Churn & Fraud Detection ML System *2025*

- Designed a dual-model production system deployed on AWS EC2, serving predictions via FastAPI endpoints with sub-200ms P95 latency
- Integrated EvidentlyAI drift reports and Grafana alerting for continuous monitoring; automated retraining triggers on PSI threshold breach

Real-Time Streaming Analytics Platform *2025*

- Built a Kafka → Flink → ClickHouse pipeline for real-time event analytics, processing >50,000 events/second with windowed aggregation and fault-tolerant checkpointing
- Provisioned infrastructure using Pulumi on AWS; end-to-end latency under 500ms for live dashboard updates

LLM-Powered Retrieval-Augmented Generation (RAG) Prototype *2024*

- Developed a document Q&A system using LangChain and LangGraph orchestration, Qdrant vector database for semantic retrieval, and a locally hosted LLM backend
- Instrumented with Langfuse for LLM observability, tracking latency, token usage, and retrieval quality metrics

PUBLICATIONS & PRESENTATIONS

Manuscripts in Preparation:

- Raj, N. A. (in preparation). "Decoding the Spectrum of Online Harassment: A Multi-Task Neural Approach to Sentiment, Hate and Aggression Identification." Target venue: ACL/EMNLP/COLING (NLP track). Expected submission: 2026.

Conference Participation:

- Active member of IEEE Computer Society Student Branch Chapter, CUET – Machine Learning Wing; participated in internal seminars and technical workshops on AI and data systems.

TECHNICAL SKILLS & COMPETENCIES

Programming & Query Languages:
Python (primary), SQL

Machine Learning & Deep Learning Frameworks:
PyTorch, HuggingFace Transformers, Scikit-Learn, NumPy, Pandas, OpenCV, LangChain, LangGraph

Distributed & Scalable Data Systems:
Apache Spark, Apache Kafka, Apache Flink, Apache Airflow

MLOps & Experiment Management:
MLflow, EvidentlyAI, Prometheus, Grafana, Langfuse

Cloud Platforms & Infrastructure-as-Code:
Amazon Web Services (EC2, S3, ECR, Lambda), Docker, Pulumi, Terraform

Databases & Vector Stores:
PostgreSQL, MongoDB, ClickHouse, Qdrant

API & Service Development:
FastAPI, RESTful API design, asynchronous Python (asyncio)

Development & Reproducibility:
Git, GitHub Actions (CI/CD), Linux (Ubuntu/Debian), Docker Compose

LEADERSHIP & PROFESSIONAL SERVICE

Secretary, Machine Learning Wing *CUET, 2023 – 2025*
IEEE Computer Society Student Branch Chapter, Chittagong University of Engineering and Technology

- Organised technical workshops and study groups on ML fundamentals, deep learning, and NLP for undergraduate members
- Coordinated inter-university academic activities and maintained correspondence with IEEE regional bodies

REFERENCES

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